AudioLLM-D: Fine-tuning Audio LLM for Dementia Speech

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1 Progress

1.1 Dataset

Following access approval to the gated DementiaBank corpus [2], preliminary analysis of the legacy Pitt Corpus revealed lack of audio quality. To mitigate this, we bypassed the uncleaned data in favor of the ADReSS challenge dataset [12], a curated, denoised subset of Pitt, alongside other high-quality registry sources [11; 6; 4; 8] , yielding an aggregated corpus of 432 sessions across 201 participants.

We developed a custom CHAT [13] parsing pipeline to isolate participant speech while mapping critical acoustic markers (pauses, fillers, repetitions, unclear speech) to dedicated tokens for representation learning. To satisfy the audio encoder's strict 30-second input constraint, files were chunked by concatenating sequential utterances up to the threshold without truncation. This pipeline produced 826 paired samples, partitioned into an 80/20 train/test split via a strict speaker-disjoint constraint to eliminate data leakage.

1.2 Training

Prior to fine-tuning, the base Qwen2-Audio-7B-Instruct model [3] was evaluated on the test split under zero-shot (system prompt defining the disfluency vocabulary) and few-shot (augmented with four synthetic exemplars) conditions. Performance was quantified via Word Error Rate (WER) and disfluency-specific classification metrics (Precision, Recall, F_1-score). A planned baseline comparison against WhisperD [1] was precluded due to its public unavailability and a lack of author response. We then fine-tuned the architecture using LoRA [7] ($r = 16$) on attention projection layers across both audio and language modules. To facilitate domain-specific disfluency learning, the newly initialized token embeddings and multimodal projector weights were also fine-tuned while all other backbone parameters remained frozen. The total parameters being fine-tuned was around 50M.

2 Challenges

Preliminary experiments reveal an optimization mismatch: token-level cross-entropy (CE) loss plateaus while test-set Word Error Rate (WER) continues to decrease, indicating CE is a poor proxy for sequence-level performance. This dynamics occurs despite our use of a weighted CE loss to counteract severe class imbalance, as the newly introduced disfluency tokens comprise only 10% of the training corpus. Furthermore, our current F_1 evaluation is order-unaware, treating disfluency detection as a "bag-of-tokens" problem. By failing to validate exact temporal placement within the transcript, this metric might overestimate model utility and limits clinical interpretability.

3 Next Steps

To optimize performance over the remaining timeline, we plan four concrete tasks. First, we will introduce per-parameter-group differential learning rates: a higher rate ($\sim 10^{-3}$) for the newly initialized disfluency token embeddings and corresponding output projection rows, standard rates ($\sim 10^{-4}$) for the language model LoRA adapters, and a lower rate ($\sim 5 \times 10^{-5}$) for the audio encoder LoRA layers to preserve pretrained acoustic features. Second, we will conduct a hyperparameter sweep over LoRA and DoRA [10] across ranks $r \in \{8, 16, 32, 64\}$; DoRA's magnitude-direction decomposition is hypothesized to better stabilize low-rank adaptation here. Third, to mitigate severe class imbalance more dynamically than standard reweighted CE, we will substitute the token objective with Focal Loss [9]. Finally, if time permits, we will explore an auxiliary Connectionist Temporal Classification head [5] on the audio encoder to provide explicit, frame-level alignment supervision for pauses and disfluent intervals [14].

References

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4 Appendix

Table 1: Overall comparison on Test Set: Zero-Shot vs Few-Shot vs LoRA fine-tuning

Scenario	WER	Token Precision	Token Recall	Token F1	TP	Pred Total	Ref Total
Zero-shot	0.5396	0.3871	0.0173	0.0331	12	31	693
Few-shot	0.6634	0.0436	0.0216	0.0289	15	344	693
LoRA fine-tuning	0.3692	0.5287	0.4921	0.5097	341	645	693

Table 2: Token-level disfluency comparison on Test Set

Token	Zero F1	Few F1	LoRA F1	Zero Prec.	Few Prec.	LoRA Prec.	Zero Rec.	Few Rec.	LoRA Rec.	ΔF1 (L−Z)	ΔF1 (L−F)
[uh]	0.0000	0.0121	0.7575	0.0000	1.0000	0.8321	0.0000	0.0061	0.6951	+0.7575	+0.7454
[um]	0.0000	0.0221	0.7500	0.0000	0.0155	0.7079	0.0000	0.0380	0.7975	+0.7500	+0.7279
[short-pause]	0.0727	0.0585	0.4172	0.8000	0.0758	0.5862	0.0381	0.0476	0.3238	+0.3445	+0.3587
[medium-pause]	0.0357	0.0000	0.2632	0.3333	0.0000	0.4348	0.0189	0.0000	0.1887	+0.2274	+0.2632
[long-pause]	0.0594	0.0138	0.5570	0.6000	0.0204	0.7097	0.0312	0.0104	0.4583	+0.4976	+0.5432
[unclear-speech]	0.0149	0.0420	0.3161	0.5000	0.2727	0.2402	0.0076	0.0227	0.4621	+0.3011	+0.2741
[laughter]	0.1111	0.0690	0.2917	0.2143	0.1111	0.8750	0.0750	0.0500	0.1750	+0.1806	+0.2227
[non-verbal]	0.0000	0.0000	0.4211	0.0000	0.0000	0.5714	0.0000	0.0000	0.3333	+0.4211	+0.4211